

# A Rigorous Reduction for the Two-Player $3n \pm 1$ Game

## Verified Lemmas, a Conditional No-Draw Theorem, and the Remaining Refinement Problem

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**Scope notice.** This manuscript does *not* claim an unconditional solution of the game. It gives complete proofs of the exact binary normal form, the outcome lemmas, the finite-token rank, and the abstract no-draw criterion. The application of that criterion to every legal game continuation still requires a concrete semantic refinement. One unresolved token-provenance defect in the current assembly, and a second defect in the earlier v6.3 router, are identified in Section 8. Consequently, the assertion that optimal play always terminates is marked OPEN TARGET, NOT CLAIMED, not PROVED. Within this stated reduction-and-audit scope, every theorem asserted by the manuscript has a proof and the final adversarial audit reports no red or orange defect.

### Abstract

From an odd integer  $n > 1$ , a player chooses  $3n + 1$  or  $3n - 1$  and removes all powers of two; reaching 1 wins. Arbitrary play may cycle, so the problem is to exclude game-theoretic draws, not merely to find a decreasing move. We derive a self-contained conjugacy to the binary game

$$q \mapsto A(q) = \left\lceil \frac{3q}{2} \right\rceil, \quad q \mapsto B(q) = R(A(q)),$$

where  $R$  deletes the maximal alternating binary suffix and  $B(q) < q$  for  $q > 0$ . We prove the exact finite-outcome recursion without imposing a finite search boundary, give a Gray-code description of  $R$ , and prove a well-founded multiset rank for finite proof witnesses. We then state and prove an explicit semantic-refinement theorem: a productive lift of every draw into a certified lexicographically decreasing macro system rules out all draws. The repository's routing certificate checks only the abstract macro system. It does not construct the required lift and successor map. A hostile audit exposes an uninstalled-witness error in the current high-return rank. It also records an outcome-orientation error in the superseded v6.3 loss-sibling router, so that the history of the argument is not silently rewritten. These are gaps in proposed global bridges, not counterexamples to the game theorem. The result of this paper is therefore a rigorous reduction with a precise open obligation. The shortest marked return is sharpened further: its apparent valuation-one subcase is empty, so that row always replaces an already retained losing token by one of its actual children.

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## 1 Claim ledger and scope

Every theorem or lemma in this paper followed by a proof is labelled PROVED. Finite experiments are labelled VERIFIED BY COMPUTATION. The global target is not inferred from either kind of finite computation.

| Claim                                                               | Status                   |
|---------------------------------------------------------------------|--------------------------|
| Exact $m$ -space and $q$ -space normal forms                        | PROVED                   |
| $B(q) < q$ and $B(B(A(q))) < q$ for $q > 0$                         | PROVED                   |
| Finite WIN/LOSS recursion and the draw-child lemma                  | PROVED                   |
| Gray-code formula for the alternating-suffix deletion               | PROVED                   |
| Finite-token replacement strictly lowers the rank                   | PROVED                   |
| Complete semantic macro refinement implies no draws                 | CONDITIONAL THEOREM      |
| The checked JSON inventory supplies that refinement                 | OPEN TARGET, NOT CLAIMED |
| Optimal play terminates from every odd start                        | OPEN TARGET, NOT CLAIMED |
| Finite identities and regression suites listed in Section 9         | VERIFIED BY COMPUTATION  |
| Fixed-tail loss-sibling and source-zero $j = 1$ local repairs       | PROVED                   |
| Shortest marked return ( $D = 1$ ) strictly pays its retained token | PROVED                   |

The distinction in the last four rows is the central audit boundary. A certificate may correctly prove that a declared transition system is well-founded while failing to prove that every relevant game situation has a legal, outcome-compatible transition in that system.

## 2 The game and its infinite-graph semantics

For a positive even integer  $x$ , let  $v_2(x)$  be the exponent of 2 in  $x$ , and put

$$\text{odd}(x) = \frac{x}{2^{v_2(x)}}.$$

The positions are the positive odd integers. Position 1 is terminal and is losing for the player to move. For odd  $n > 1$ , the two children are

$$T_-(n) = \text{odd}(3n - 1), \quad T_+(n) = \text{odd}(3n + 1).$$

The graph is infinite and may contain cycles; for example  $5 \rightarrow 7 \rightarrow 5$  is legal. Finite-game backward induction may therefore not be used with an artificial boundary.

### 2.1 Finite proof trees

**Definition 2.1** (Finite outcome layers). Let  $L_0 = \{1\}$  and  $W_0 = \emptyset$ . Inductively, for  $h \geq 0$ , define

$$\begin{aligned} W_{h+1} &= W_h \cup \{n > 1 : \text{at least one child of } n \text{ belongs to } L_h\}, \\ L_{h+1} &= L_h \cup \{n > 1 : \text{both children of } n \text{ belong to } W_h\}. \end{aligned}$$

Put  $W = \bigcup_h W_h$ ,  $L = \bigcup_h L_h$ , and  $D = \{1, 3, 5, \dots\} \setminus (W \cup L)$ . Membership in  $W$  or  $L$  is witnessed by a finite proof tree. The least layer of membership is its proof height.

**Lemma 2.2** (Disjointness; PROVED).  $W \cap L = \emptyset$ .

*Proof.* Suppose otherwise and choose a position  $x$  with a pair of  $W$ - and  $L$ -derivations for which the sum of the two derivation heights is minimal. The terminal position is never in  $W$ , so  $x > 1$ . The  $W$ -derivation chooses a child  $y$  having an  $L$ -derivation of smaller height. The  $L$ -derivation requires every child, including  $y$ , to have a  $W$ -derivation of smaller height. Thus  $y$  has both labels with a smaller sum of heights, a contradiction.  $\square$

**Definition 2.3** (Outcomes). We call the positions in  $W$ ,  $L$ , and  $D$  respectively WIN, LOSS, and DRAW. The word “draw” means failure of both finite winning certificates. It is not the same as “not reached by a bounded search”.

**Proposition 2.4** (Exact recursion; PROVED). *For every nonterminal position  $x$ :*

- (i)  $x$  is WIN if and only if at least one child is LOSS;
- (ii)  $x$  is LOSS if and only if both children are WIN;
- (iii) if  $x$  is DRAW, then at least one child is DRAW and no child is LOSS.

*Proof.* The forward implications in (i) and (ii) follow from the last rule in a minimal finite derivation. Conversely, a finite proof for a losing child may be extended by one node to a winning proof for its parent; two finite winning proofs may be joined below a losing parent. This proves (i) and (ii). If a draw had a losing child, (i) would make it winning. If neither child were a draw, both would be resolved; since neither is losing, both would be winning, and (ii) would make the parent losing. This proves (iii).  $\square$

**Proposition 2.5** (Game-theoretic meaning; PROVED). *At a WIN position, the player to move has a strategy reaching 1 after finitely many moves. At a LOSS position, every first move gives the opponent such a strategy. At a DRAW position neither player can force a win. Hence, if  $D = \emptyset$ , optimal play terminates at 1 from every starting position.*

*Proof.* Induct on the finite proof height. At a winning node, move to the certified losing child. At a losing node, either move leads to a winning node of lower proof height in its attached finite proof tree. Along play consistent with the relevant winning strategy the heights decrease through a finite tree, so the terminal node is reached.

Now start at a draw. Proposition 2.4(iii) supplies a draw child and excludes losing children. Against any proposed winning strategy, the opponent preserves a draw whenever the proposed strategy leaves a draw; if it instead moves to a winning child (winning for the new player to move), the opponent switches to that child's finite winning strategy. Thus the proposed strategy cannot force a win. The same argument applies to either player. The final assertion follows from Lemma 2.2 and the definition of  $D$ .  $\square$

### 3 Exact binary normal form

Write an odd position as  $n = 2m + 1$ ,  $m \geq 0$ , and define

$$F(m) = \left\lceil \frac{3m}{2} \right\rceil.$$

For  $m > 0$ , let  $R(m)$  be the integer obtained by deleting the maximal alternating suffix of the ordinary binary expansion of  $m$ . If the whole word is alternating, the result is 0; set  $R(0) = 0$ . For example,

$$R(1001_2) = 10_2, \quad R(10101_2) = 0.$$

The following recurrence is an exact definition suitable for induction:

$$\begin{aligned} R(2a) &= \begin{cases} a, & a \text{ even,} \\ R(a), & a \text{ odd,} \end{cases} \\ R(2a + 1) &= \begin{cases} R(a), & a \text{ even,} \\ a, & a \text{ odd.} \end{cases} \end{aligned} \tag{1}$$

Indeed, the low bit extends the alternating suffix exactly when it differs from the preceding bit.

**Theorem 3.1** ( $m$ -space normal form; PROVED). *For  $n = 2m + 1 > 1$ , the two children, expressed again as  $n' = 2m' + 1$ , have coordinates*

$$\boxed{m' = F(m) \quad \text{and} \quad m' = F(R(m)).}$$

*The unordered pair is meant; the sign producing each member depends on the parity of  $m$ .*

*Proof.* Since

$$3n - 1 = 2(3m + 1), \quad 3n + 1 = 2(3m + 2),$$

exactly one of  $3m + 1$  and  $3m + 2$  is odd. If  $m = 2a$ , the odd expression is  $6a + 1$  and its new coordinate is  $3a = F(m)$ . If  $m = 2a + 1$ , the odd expression is  $6a + 5$  and its new coordinate is  $3a + 2 = F(m)$ .

It remains to identify the branch whose inner expression is even. We use strong induction on  $m$ . For  $m = 0$  the only relevant position is terminal. If  $m = 2a$ , the other odd child is  $\text{odd}(3a + 1)$ . When  $a$  is even,  $3a + 1$  is odd, its coordinate is  $3a/2 = F(a)$ , and (1) gives  $R(m) = a$ . When  $a$  is odd,  $3a + 1$  is the even branch occurring in the smaller input  $a$ ; the induction hypothesis gives coordinate  $F(R(a))$ , and  $R(m) = R(a)$ . If  $m = 2a + 1$ , the other odd child is  $\text{odd}(3a + 2)$ . When  $a$  is odd it has coordinate  $(3a + 1)/2 = F(a)$  and  $R(m) = a$ ; when  $a$  is even, the case  $a = 0$  gives coordinate  $0 = F(R(0))$  directly, while for  $a > 0$  the induction hypothesis at  $a$  gives  $F(R(a))$  and  $R(m) = R(a)$ . These four cases prove the formula.  $\square$

### 3.1 Conjugation

The image of  $F$  is precisely the set of nonnegative integers not congruent to 1 (mod 3), and

$$F^{-1}(y) = \begin{cases} 2y/3, & y \equiv 0 \pmod{3}, \\ (2y - 1)/3, & y \equiv 2 \pmod{3}. \end{cases} \quad (2)$$

This follows immediately by separating even and odd inputs of  $F$ . Every child in Theorem 3.1 lies in this image. Represent the  $m$ -coordinate  $F(q)$  simply by  $q$ .

**Theorem 3.2** (Conjugated game; PROVED). *After the first move, the game is exactly the game on  $q \in \mathbb{N}$  with terminal state 0 and moves*

$$\boxed{A(q) = F(q) = \left\lceil \frac{3q}{2} \right\rceil, \quad B(q) = R(F(q)).}$$

Moreover, resolving every  $q$ -position resolves every original odd position.

*Proof.* Apply Theorem 3.1 to a state whose  $m$ -coordinate is  $F(q)$ . Its child coordinates are  $F(F(q))$  and  $F(R(F(q)))$ ; after applying  $F^{-1}$  these are  $A(q)$  and  $B(q)$ . The terminal coordinate is  $0 = F(0)$ . For an arbitrary original state with coordinate  $m$ , its two children have coordinates  $F(m)$  and  $F(R(m))$ , so they are represented by  $m$  and  $R(m)$ . If every represented position is resolved, both children of the original state are resolved, and Proposition 2.4 resolves the parent.  $\square$

**Lemma 3.3** (Strict contraction of  $B$ ; PROVED). *For every  $q > 0$ ,  $B(q) < q$ .*

*Proof.* Deleting a nonempty suffix gives  $R(x) \leq \lfloor x/2 \rfloor$  for  $x > 0$ . Thus, for  $q \geq 2$ ,

$$B(q) \leq \left\lfloor \frac{F(q)}{2} \right\rfloor \leq \left\lfloor \frac{3q + 1}{4} \right\rfloor < q.$$

Directly,  $A(1) = 2 = 10_2$  and  $B(1) = 0$ .  $\square$

**Lemma 3.4** (A short descending block; PROVED). *For every  $q > 0$ ,*

$$B(B(A(q))) < q,$$

*where a move reaching 0 is left at 0.*

*Proof.* From the proof of Lemma 3.3,  $B(x) \leq (3x + 1)/4$  and  $A(q) \leq (3q + 1)/2$ . Therefore

$$B(B(A(q))) \leq \frac{27q + 23}{32} < q$$

for  $q \geq 5$ . For  $q = 1, 2, 3, 4$ , direct binary suffix deletion gives the respective values 0, 0, 1, 1, all strictly below  $q$ .  $\square$

**Remark 3.5** (Why this is not a termination proof). Lemma 3.4 controls a path segment containing two specified contracting moves. It says nothing about arbitrarily long runs of  $A$ -moves, and it does not determine which child is compatible with DRAW. Therefore it cannot by itself exclude an infinite optimal-play path.

## 4 The suffix operation in valuation and Gray coordinates

**Lemma 4.1** (Valuation formula; PROVED). *For every  $x > 0$ ,*

$$R(x) = \begin{cases} \lfloor x/2^{v_2(3x+1)} \rfloor, & x \text{ odd}, \\ \lfloor x/2^{v_2(3x+2)} \rfloor, & x \text{ even}. \end{cases} \quad (3)$$

*Proof.* Let the maximal alternating suffix have length  $k$ . If  $x$  is odd, its low bits begin  $1, 0, 1, 0, \dots$ ; if  $x$  is even, they begin  $0, 1, 0, 1, \dots$ . The finite geometric sums give, modulo  $2^k$ ,

$$x \equiv -3^{-1} \pmod{2^k} \quad (x \text{ odd}), \quad x \equiv -2 \cdot 3^{-1} \pmod{2^k} \quad (x \text{ even}).$$

Thus  $2^k$  divides  $3x + 1$  in the odd case and  $3x + 2$  in the even case. If a nonempty prefix remains, the boundary bit repeats the top suffix bit; using the same geometric sum modulo  $2^{k+1}$  shows that divisibility by  $2^{k+1}$  fails. Hence the displayed valuation equals  $k$ , and division by that power of two deletes exactly the suffix. If the suffix is the entire binary word, the valuation may exceed  $k$ , but both the quotient's floor and  $R(x)$  are 0. This proves (3).  $\square$

**Lemma 4.2** (No fixed modulus; PROVED). *For every  $K \geq 1$  there exist  $x \equiv y \pmod{2^K}$  with  $R(x) \neq R(y)$ . Consequently, no fixed residue modulo  $2^K$  determines  $R$  globally.*

*Proof.* Fix the same alternating word, with low bit 1, in the lowest  $K$  positions of  $x$  and  $y$ . For  $x$ , choose the bit in position  $K$  to repeat the bit in position  $K - 1$ ; for  $y$ , choose it to be the opposite bit and choose the next bit to repeat it. Add one further leading 1 if either finite word would otherwise have a leading zero. Then the two integers agree in their lowest  $K$  bits. The maximal alternating suffix of  $x$  has length  $K$ , whereas that of  $y$  has length  $K + 1$ . After deletion, the low bit of  $R(x)$  is the bit in position  $K$ , while the low bit of  $R(y)$  is the opposite bit in position  $K + 1$ . The retained prefixes are nonzero after the possible leading-bit addition; their parities differ, so  $R(x) \neq R(y)$ .  $\square$

Let

$$G(x) = x \text{ xor } \left\lfloor \frac{x}{2} \right\rfloor$$

be reflected binary Gray code.

**Theorem 4.3** (Gray deletion formula; PROVED). *For every  $x \geq 0$ ,*

$$\boxed{G(R(x)) = \left\lfloor \frac{G(x)}{2^{v_2(G(x)+1)+1}} \right\rfloor.}$$

*Proof.* Write the binary digits of  $x$  as  $b_i$ . Away from the leading boundary, the Gray digit in position  $i$  is  $b_{i+1} \text{ xor } b_i$ . If the maximal alternating suffix has length  $k$  and a nonempty prefix remains, the lowest  $k - 1$  Gray digits are 1 and the boundary Gray digit is 0. Thus  $v_2(G(x) + 1) = k - 1$ , and shifting by  $k$  leaves exactly the Gray code of the remaining prefix. If the entire word is alternating,  $G(x)$  is an all-ones word; the displayed shift removes the whole word (and one leading zero), so both sides are 0. The case  $x = 0$  is immediate.  $\square$

**Proposition 4.4** (Eight-state transducer; PROVED). *Read Gray words from least significant to most significant, with leading-zero padding. The relation*

$$h = G(A(G^{-1}(g)))$$

*is recognized by a finite transducer with eight carry states.*

*Proof.* Let  $b_i, c_i$  be the binary digits of  $q, A(q)$  and let  $g_i = b_{i+1} \text{ xor } b_i$ ,  $h_i = c_{i+1} \text{ xor } c_i$  be their Gray digits. Since

$$2A(q) = 3q + b_0 = q + 2q + b_0,$$

a state at column  $i$  may be taken as  $(b_i, c_i, d_{i+1}) \in \{0, 1\}^3$ , where  $d_{i+1}$  is the carry into column  $i + 1$ . On reading  $(g_i, h_i)$ , set

$$b_{i+1} = b_i \text{ xor } g_i, \quad c_{i+1} = c_i \text{ xor } h_i.$$

The transition is legal exactly when

$$c_i \equiv b_{i+1} + b_i + d_{i+1} \pmod{2}, \quad d_{i+2} = \left\lfloor \frac{b_{i+1} + b_i + d_{i+1}}{2} \right\rfloor.$$

Initially  $d_1 = b_0$ ; the bits  $b_0, c_0$  are guessed. Accept after zero padding reaches  $(0, 0, 0)$ . Gray inversion with the leading-zero condition makes both guesses unique. The displayed column equations then hold if and only if  $2A(q) = 3q + b_0$ , proving that the accepted pairs are exactly the stated relation. There are  $2^3 = 8$  states.  $\square$

Together with Theorem 4.3, this proves that both game edges are rational binary relations. Finite-state representability, however, is not a well-founded rank and does not exclude draws.

## 5 Finite proof witnesses and a sound token rank

Suppose a local outcome argument uses already certified finite proof trees. Record only their heights in a finite multiset  $M$ . The word “already” is essential: a finite state discovered later is not automatically a token carried by the current configuration.

Define the numerical rank

$$\tau(M) = \sum_{h \in M} 3^h,$$

with multiplicity. This is a dependency-free substitute for the natural ordinal sum of  $\omega^h$  terms when every replacement has at most two children.

**Lemma 5.1** (Token replacement; PROVED). *Let  $M = U \uplus \{h\}$ , and let  $M' = U \uplus \{h_1, \dots, h_k\}$ , where  $0 \leq k \leq 2$  and every  $h_i < h$ . Then  $\tau(M') < \tau(M)$ .*

*Proof.* If  $k = 0$  the claim is immediate. Otherwise  $h \geq 1$  and

$$\sum_{i=1}^k 3^{h_i} \leq 2 \cdot 3^{h-1} < 3^h.$$

The unchanged contribution  $\tau(U)$  may be added to both sides.  $\square$

**Corollary 5.2** (Well-founded token descent; PROVED). *There is no infinite sequence of finite token multisets in which every step is a replacement of the form in Lemma 5.1.*

*Proof.* The natural number  $\tau(M)$  strictly decreases at every step.  $\square$

**Remark 5.3** (Provenance rule). Lemma 5.1 permits replacing a particular retained occurrence  $h \in M$ . It does not permit choosing an unrelated finite proof of height  $u$ , proving that new heights are below  $u$ , and inserting them into  $M$  when  $u$  was not in  $M$ . For example, if  $M = \emptyset$ , inserting any nonempty list increases  $\tau$ , regardless of comparisons with witnesses external to  $M$ .

## 6 Three local router repairs proved explicitly

This section proves two draw-preserving constructions that were only named, but not constructed, in the earlier v6.3 manuscript: the fixed-tail loss-sibling orientation and the source-zero valuation-one exit. It also proves that the formally possible valuation-one branch of the shortest marked return is empty. None of these local lemmas repairs the separate long high-return token-installation defect of Section 8.

For  $r \geq 1$ ,  $e \in \{0, 1\}$ , and odd  $a > 0$ , put

$$Q_r^e(a) = 2^r a - e, \quad J(s) = 2A(s) + 1.$$

Thus

$$J(s) = \begin{cases} 3s + 1, & s \text{ even}, \\ 3s + 2, & s \text{ odd}. \end{cases}$$

**Lemma 6.1** (Canonical coefficient source; PROVED). *Every  $x > 0$  has unique coordinates*

$$x = Q_r^e(3^k J(s)), \quad r \geq 1, \quad e \in \{0, 1\}, \quad k, s \in \mathbb{N}.$$

If  $\sigma(x) = s$  denotes the resulting coefficient source, then

$$\sigma(x) \leq \frac{x - 1}{6}.$$

*Proof.* The parity of  $x$  determines  $e$ : namely  $e = x \bmod 2$ . The factorization  $x + e = 2^r a$  with  $a$  odd determines  $r \geq 1$  and  $a$  uniquely. Removing the maximal power  $3^k$  determines an odd integer  $a_0$  not divisible by three. Such integers are precisely  $6t + 1$  and  $6t + 5$ . The displayed formula for  $J$  gives

$$J(2t) = 6t + 1, \quad J(2t + 1) = 6t + 5,$$

so  $a_0 = J(s)$  for a unique  $s$ . Finally

$$x = 2^r 3^k J(s) - e \geq 2J(s) - 1 \geq 6s + 1,$$

which is the asserted bound. □

Call  $q$  *exceptional* when

$$q \equiv 1, 3, 12, 14 \pmod{16};$$

all other  $q$  are called *ordinary*.

**Lemma 6.2** (Ordinary side relation; PROVED). *If  $q$  is ordinary, then*

$$B(A(q)) \in \{A(B(q)), B(B(q))\}.$$

*Moreover, if  $q$  is exceptional, then  $A(q)$  is ordinary.*



*Proof.* Put  $x = A(q)$ ,  $s = R(x) = B(q)$ , and let  $k$  be the length of the maximal alternating suffix of  $x$ . The condition  $k \geq 3$  says that the last three bits of  $x$  are 010 or 101. Substitution of the sixteen residues of  $q$  into  $A(q) = \lceil 3q/2 \rceil$  shows that this occurs exactly for  $q \equiv 1, 3, 12, 14 \pmod{16}$ . Thus  $k \in \{1, 2\}$  when  $q$  is ordinary.

If  $k = 1$ , write  $x = 2s + e$ , where the boundary condition for maximality says  $s \equiv e \pmod{2}$ . The formulas

$$A(2s) = 3s, \quad A(2s + 1) = 3s + 2$$

show that  $A(x)$  is obtained from  $A(s)$  by appending the bit  $e$ . Deleting the new maximal alternating suffix therefore leaves  $A(s)$  if the boundary bit repeats and leaves  $R(A(s)) = B(s)$  if it continues.

If  $k = 2$ , the two deleted bits are 01 or 10, and their high bit equals the last bit of  $s$ . Directly,  $x = 4s + 1$  with  $s$  even gives  $A(x) = 4A(s) + 2$ , while  $x = 4s + 2$  with  $s$  odd gives  $A(x) = 4A(s) + 1$ . Thus  $A(x)$  is obtained from  $A(s)$  by appending, respectively, 10 or 01. The same boundary test leaves either  $A(s)$  or  $R(A(s)) = B(s)$ . This proves the first assertion. For the second, direct substitution sends the four exceptional classes under  $A$  to 2, 5, 10, 13  $\pmod{16}$ , none of which is exceptional.  $\square$

**Lemma 6.3** (Exceptional child-pair attachment; PROVED). *Let  $q$  be exceptional and put  $\ell = B(q)$ . For some  $m \geq 1$  and  $\delta \in \{0, 1\}$ , the two children of  $A(q)$  are exactly*

$$Q_m^\delta(J(\ell)), \quad Q_{m+1}^\delta(J(\ell)).$$

*Consequently, if  $q$  is WIN,  $A(q)$  is DRAW, and  $\ell$  is LOSS, then this exact adjacent pair contains a continuing DRAW and remains attached to the same losing state  $\ell$ .*

*Proof.* Write  $q = 16t + c$ ,  $c \in \{1, 3, 12, 14\}$ . Direct application of  $A$  and  $B$  gives

| $c$ | $\ell$      | $B(A(q))$  | $A^2(q)$     |
|-----|-------------|------------|--------------|
| 1   | $R(6t)$     | $18t + 1$  | $36t + 3$    |
| 3   | $R(6t + 1)$ | $18t + 4$  | $36t + 8$    |
| 12  | $R(6t + 4)$ | $18t + 13$ | $36t + 27$   |
| 14  | $R(6t + 5)$ | $18t + 16$ | $36t + 32$ . |

In the four rows put  $z = 6t, 6t + 1, 6t + 4, 6t + 5$ , respectively. If the maximal alternating suffix of  $z$  has length  $m$  and leaves  $\ell = R(z) > 0$ , direct summation of that alternating suffix gives

$$3z + 1 = Q_m^\delta(J(\ell)), \quad \delta = 1 - (z \bmod 2).$$

When the whole word is deleted, the same identity holds with  $\ell = 0$  and  $m$  one larger than the binary length. The third column is  $3z + 1$  and the fourth is twice that number plus  $\delta$ , proving the two coordinates. If  $A(q)$  is a draw, its children contain no loss and at least one draw by Proposition 2.4; the final assertion follows.  $\square$

**Lemma 6.4** (Fixed-tail loss-sibling constructor; PROVED). *Let  $c > 0$  be odd,  $r \geq 2$ , and  $e \in \{0, 1\}$ . Put*

$$X = Q_r^e(3c), \quad G = Q_r^e(c), \quad H = Q_{r+1}^e(c), \quad Y = A(G).$$

*Then  $\{A(H), B(H)\} = \{X, Y\}$ , in fact  $A(H) = X$  and  $B(H) = Y$ . Suppose*

$$X : \text{DRAW}, \quad H : \text{WIN}, \quad Y : \text{LOSS}.$$

*There is an explicit draw-preserving continuation:*

(a) if  $H$  is ordinary, then  $P = B(A(H))$  is simultaneously a child of  $X$  and of  $Y$ ; hence

$$P : \text{WIN}, \quad X \rightarrow P \text{ is a draw-win boundary}, \quad h(P) \leq h(H) - 2;$$

(b) if  $H$  is exceptional, the two children of the actual draw  $X = A(H)$  are the exact adjacent pair of Lemma 6.3 over the same losing state  $Y = B(H)$ , and that pair contains a draw.

No transition through the winning state  $G$  is asserted.

*Proof.* For  $r \geq 3$ , constant-tail arithmetic gives

$$A(H) = Q_r^e(3c) = X, \quad B(H) = Q_{r-1}^e(3c) = A(G) = Y.$$

At  $r = 2$  the same second identity is the direct exponent-two/one boundary calculation; the first identity is unchanged.

Assume first that  $H$  is ordinary. Lemma 6.2 gives

$$P = B(A(H)) \in \{A(B(H)), B(B(H))\} = \{A(Y), B(Y)\}.$$

Thus  $P$  is a child of the losing state  $Y$  and is winning. It is also the contracting child of  $A(H) = X$ , so it is a genuine winning child of the continuing draw. Since a draw cannot be the losing witness for the winning state  $H$ , the stated orientation makes  $Y$  its losing witness and

$$h(Y) = h(H) - 1, \quad h(P) \leq h(Y) - 1 = h(H) - 2.$$

If  $H$  is exceptional, apply Lemma 6.3 with  $q = H$ ,  $A(H) = X$ , and  $B(H) = Y$ . This proves the second alternative and the exhaustiveness of the split.  $\square$

**Lemma 6.5** (Source-zero valuation-one exit; PROVED). *Let  $n \geq 2$ ,  $\delta \in \{0, 1\}$ , and suppose*

$$3^n + 1 - 2\delta = 2J(y)$$

*has 2-adic valuation one. Put*

$$C = R(3^n - \delta), \quad E = Q_1^{1-\delta}(J(y)).$$

*Then  $y = (3^{n-1} - 1)/2 > 0$  and exactly one of the following two coordinate rows holds:*

| $\delta$ | $n$  | $C$    | $E$           |
|----------|------|--------|---------------|
| 0        | even | $A(y)$ | $Q_1^1(J(y))$ |
| 1        | odd  | $B(y)$ | $Q_1^0(J(y))$ |

*If  $\{C, E\}$  contains a draw, then either  $E$  itself is an explicit exponent-one draw lift over  $y$ , or  $C$  is a draw with*

$$\sigma(C) < y.$$

*Thus the raw valuation-one row is a strict coefficient-source exit, not an unconstructed loss-sibling entry.*

*Proof.* Valuation one in  $3^n + 1$  forces  $n$  even, while valuation one in  $3^n - 1$  forces  $n$  odd; this follows modulo four. Dividing the displayed factorization by two gives  $J(y) = (3^n + 1 - 2\delta)/2$ , and the explicit formula for  $J$  yields  $y = (3^{n-1} - 1)/2$ . Direct substitution gives the two signed coordinates. In the first row the last two bits of  $3^n$  are 01, so deletion gives  $C = (3^n - 1)/4 = A(y)$ . In the second row appending the final zero to  $J(y)$  extends the alternating suffix of  $A(y)$ , giving  $C = B(y)$ .

If  $E$  is a draw, the first alternative is already explicit. Otherwise a draw in the pair must be  $C$ . Lemma 6.1 and  $A(y) \leq (3y + 1)/2$  give

$$\sigma(A(y)) \leq \frac{A(y) - 1}{6} \leq \frac{3y - 1}{12} < y.$$

For the other row, Lemmas 3.3 and 6.1 give  $\sigma(B(y)) < y$ . Hence both raw rows make the claimed strict source descent.  $\square$

**Lemma 6.6** (Signed source boundary; PROVED). *Let  $s > 0$ ,  $e \in \{0, 1\}$ , and factor*

$$3J(s) + 1 - 2e = 2^v J(t).$$

*Then*

$$t = \begin{cases} A(s), & v = 1, \\ B(s), & v \geq 2. \end{cases}$$

*Proof.* The odd integer  $J(s)$  is the original game state represented by  $s$ . Theorem 3.2 says that the sources of its two signed children are exactly  $A(s)$  and  $B(s)$ . Of the consecutive even integers  $3J(s) - 1$  and  $3J(s) + 1$ , one has valuation one and the other has valuation at least two. The valuation-one odd quotient is greater than  $J(s)$ , whereas the valuation-at-least-two quotient is smaller than  $J(s)$ . Since  $J$  is strictly increasing,  $A(s) > s > B(s)$  by Lemma 3.3; hence the first quotient has source  $A(s)$  and the second has source  $B(s)$ .  $\square$

**Lemma 6.7** (The shortest marked return is strict; PROVED). *Let  $C > 0$  be odd, let  $g \in \{0, 1\}$ , and put*

$$b = Q_1^g(C), \quad q = B(b), \quad y = A(b).$$

*Suppose that  $b, q$  are WIN and  $y$  is LOSS. Factor*

$$9J(b) + 1 - 2g = 2^j J(w).$$

*Then  $j \geq 2$  and*

$$w = \begin{cases} A(y), & j = 2, \\ B(y), & j \geq 3. \end{cases}$$

*Consequently  $w$  is an actual winning child of the retained losing state  $y$ , and*

$$h(w) \leq h(y) - 1.$$

*In particular this marked  $D = 1$  row is a legal strict token replacement; it has no valuation-one lift subcase.*

*Proof.* Because  $b = 2C - g$  has parity  $g$ , the formula for  $J$  gives

$$J(b) = 6C + 1 - 2g.$$

Hence

$$9J(b) + 1 - 2g = 2(27C + 5 - 10g).$$

The integer in parentheses is even for both values of  $g$ , since  $C$  is odd. Thus  $j \geq 2$ .

It remains to identify the returned source. First note that

$$y = A(b) = 3C - g,$$

and therefore

$$J(y) = \begin{cases} 9C + 2, & g = 0, \\ 9C - 2, & g = 1. \end{cases}$$

The remaining factorization is precisely the signed source boundary at  $y$ :

$$3J(y) + 1 - 2(1 - g) = 27C + 5 - 10g = 2^{j-1}J(w).$$

If  $j = 2$ , its valuation is one and Lemma 6.6 gives  $w = A(y)$ . If  $j \geq 3$ , its valuation is at least two and the same lemma gives  $w = B(y)$ .

In both cases  $w$  is a child of the retained LOSS state  $y$ . Proposition 2.4 makes it WIN, and adjoining this child to the finite losing proof for  $y$  gives  $h(w) \leq h(y) - 1$ .  $\square$

## 7 The abstract macro theorem

This section states exactly what a global certificate would have to prove. It deliberately separates a sound order-theoretic kernel from the semantic connection to the game.

Let a macro configuration be

$$C = (s, M, a, N, c),$$

where  $s, a \in \mathbb{N}$  are outer and inner source parameters,  $M, N$  are finite multisets of retained proof heights, and  $c$  belongs to a finite control set. Assume the equal-rank control graph is acyclic, and choose a topological rank  $d(c) \in \mathbb{N}$  that decreases on every control edge. Put

$$\rho(C) = (s, \tau(M), a, \tau(N), d(c)) \in \mathbb{N}^5$$

with lexicographic order.

**Lemma 7.1** (Macro well-foundedness; PROVED). *Any relation  $C' \prec C$  satisfying  $\rho(C') <_{\text{lex}} \rho(C)$  is well-founded.*

*Proof.* The usual order on  $\mathbb{N}$  is well-founded, and a finite lexicographic product of well-founded orders is well-founded. Concretely, an infinite descending sequence would first strictly decrease the first coordinate only finitely often. On a tail it is constant; repeat this argument successively for the remaining four coordinates. The resulting tail would have all coordinates constant, contradicting strict lexicographic decrease.  $\square$

**Definition 7.2** (Complete draw-macro refinement). A complete refinement consists of a predicate  $\text{Bad}(C)$  and two constructions:

- (R1) for every conjugated draw  $q$ , a configuration  $C$  with  $\text{Bad}(C)$ ;
- (R2) for every  $C$  with  $\text{Bad}(C)$ , a configuration  $C'$  with  $\text{Bad}(C')$  and  $C' \prec C$ .

Here every macro edge used in (R2) must be a finite legal game construction, must follow an actual DRAW continuation supplied by Proposition 2.4, and must preserve the identities and provenance of all tokens unless Lemma 5.1 certifies their strict replacement.

**Theorem 7.3** (Conditional no-draw theorem; PROVED). *If a complete draw-macro refinement in the sense of Definition 7.2 exists, then the conjugated game has no draw positions. Consequently, optimal play in the original game terminates at 1 from every odd starting value.*

*Proof.* If a draw existed, (R1) would make the set of bad configurations nonempty. By Lemma 7.1, every nonempty subset has a  $\prec$ -minimal element  $C$ . But (R2) supplies a bad  $C' \prec C$ , a contradiction. Hence there are no conjugated draws. Theorem 3.2 resolves every original position, and Proposition 2.5 gives finite termination under optimal play.  $\square$

## 7.1 What the routing certificate does and does not prove

The checked inventory has 15 control states, 45 declared transitions, 12 declared total guard partitions, 29 rank-strict transitions, and 16 equal-rank transitions forming a directed acyclic graph. These facts prove Lemma 7.1 for the *declared* macro relation. To instantiate Theorem 7.3, four additional obligations must be proved:

- (H1) **Arithmetic validity.** Every declared row has the asserted exact child, source, exponent, and suffix identities.
- (H2) **Semantic coverage.** The guards partition every game situation that can arise from a draw; there is no missing entry or reset case.
- (H3) **Outcome and token compatibility.** The selected route follows an actual draw child, and every rank comparison concerns the actual retained token occurrences, not newly introduced finite witnesses.
- (H4) **Finite productivity.** Each macro invocation reaches a declared successor after finitely many legal constructions, and a reset occurs only after a preceding rank component has strictly decreased.

The JSON checker validates declarations about the abstract inventory. It cannot establish (H2)–(H4) merely by confirming that the declared guards are syntactically disjoint and exhaustive over their own declared parameter domains. Likewise, the Lean structure called `DrawMacroRefinement` makes (R1)–(R2) explicit hypotheses; no concrete value of that structure is currently constructed.

## 8 Hostile audit of the proposed global bridge

This section records proof defects, not merely requests for more exposition. Defect A applies to the current human assembly. Defect B was a defect of the earlier v6.3 manuscript; it is repaired explicitly by Lemma 6.4. The source-zero valuation-one omission from that manuscript is repaired by Lemma 6.5. The current proof still cannot instantiate Theorem 7.3 until Defect A and the explicit semantic-refinement boundary are closed.

### 8.1 Defect A: the high-return witnesses are not installed tokens

In the high-return module, the argument constructs finite winning states  $u, c$  and lower descendants  $p, b$ , with height inequalities of the schematic form

$$h(p) < h(u), \quad h(b) < h(c).$$

The next assembly step treats this as a strict replacement

$$\{h(u), h(c)\} \mapsto \{h(p), h(b)\}$$

inside the current token multiset. However, the local lemma proves only that  $u$  and  $c$  are finite winning states; it does not prove that occurrences of  $h(u)$  and  $h(c)$  were already retained in the current outer or inner multiset.

This is a logical failure of the premise of Lemma 5.1. If the current multiset is empty, for example, installing  $h(p)$  and  $h(b)$  makes  $\tau$  positive rather than smaller. Comparisons with external witnesses do not repair the problem. A valid repair must either:

- (a) trace  $u$  and  $c$  as exact retained token occurrences from the entry configuration to the high return; or
- (b) define a different rank that pays for their installation before the return and prove strict descent for every reset.

No such trace or prepaid rank is currently supplied. Therefore the claimed strict high-return edge is OPEN TARGET, NOT CLAIMED.

## 8.2 Repaired defect B (v6.3 only): the loss-sibling orientation

One entry pattern in the v6.3 normalizer has the outcome fingerprint

$$X : \text{DRAW}, \quad H : \text{WIN}, \quad Y : \text{LOSS}, \quad G : \text{WIN}.$$

The available local diamond keeps the draw at  $X$ . The assembly nevertheless routes the configuration through a “loss-sibling entry” based at  $G$ . Because  $G$  is winning rather than drawing, the v6.3 transition was not an outcome-compatible continuation of  $X$ . Lemma 6.4 now supplies the missing constructor without moving the draw to  $G$ . In the ordinary row, the actual draw  $X$  has a winning child two proof levels below  $H$ ; in the exceptional row, the two actual children of  $X$  form an exact adjacent frame over the same losing state  $Y$ . Thus this historical defect is closed in the present manuscript.

## 8.3 Additional obligations exposed by the audit

Two further points in v6.3 required explicit case proofs in addition to Defect B:

- the  $D = 2$  loss-anchored lift/factor exits must preserve or lower the exact incoming token in every signed and raw exit, not merely produce some finite witness; this local obligation is now closed by the exact split  $B(w) \in \{A^2(q), B(A(q))\}$  and the ordinary/exceptional common-child identities recorded in the repository note *D2 loss-anchored obligation*, conditional only on the incoming occurrence of  $q$  being legally installed;
- the source-zero, valuation-one branch required an explicit route; Lemma 6.5 now proves that the signed exit is an exact exponent-one lift and the raw exit is a strict coefficient-source descent, so no new loss-sibling constructor is needed.

None of these observations disproves the desired game theorem. Defect A shows that the current token installation and lifecycle have not yet been justified. The loss-sibling, source-zero, shortest marked  $D = 1$ , and local  $D = 2$  rows are now closed. The long high-return carry/no-reseed theorem remains the explicit global obligation. Independently, the current Lean development leaves the complete draw-to-macro map as an explicit structure parameter. Thus the last implication needed for an unconditional proof is unavailable. The exact acceptance conditions for Defect A are recorded in the repository note *High-return provenance obligation*.

## 9 Computational verification and its exact scope

All commands below were run on 12 August 2026 using Python 3.12.13. They are reproducible from the repository snapshot audited for this manuscript.

```
python -m unittest discover -s tests -v
python scripts/verify_claims.py --limit 100000
python scripts/verify_global_certificate.py
python audit.py --limit 100000
```

The results were:

- 112 unit and regression tests passed;
- the  $m$ -space normal form was checked for  $1 \leq m \leq 100,000$ ;
- $B(q) < q$  was checked for  $1 \leq q \leq 100,000$ ;
- descent blocks were checked for odd  $3 \leq n \leq 200,001$ ;
- the global inventory checker accepted 15 states and 45 transitions, reporting the status `CONDITIONAL_MACHINE_CHECK` and four unverified local refinement/outcome obligations;
- an independently verified finite proof DAG certified  $q = 100$  as winning, using 59 nodes and maximum finite rank 44,448.

The previous manuscript v6.3 was also tested by

```
python verify_structure_v6_3.py
python verify_v6_3.py --limit 250000 \
--random-samples 5000 --random-bits 256
```

Both scripts passed. The second script explicitly describes its scope as “arithmetic/control regressions only; human proof carries outcomes/ranks.” This is consistent with Defects A and B: neither is a false integer identity detectable by a finite arithmetic test.

These results are VERIFIED BY COMPUTATION, not theorems about all integers. In particular, an unknown node outside a finite retrograde window is not labelled winning or losing, and no finite cutoff is treated as a boundary condition.

## 10 A precise target for future work

The open problem has now been reduced to a concrete construction rather than an informal appeal to a complicated case analysis.

**Proposition 10.1** (Sufficient remaining deliverable; PROVED). *It is sufficient to provide, in Lean or on paper, explicit functions implementing (R1)–(R2) of Definition 7.2, together with proofs of (H1)–(H4). No additional global game-theoretic argument would then be needed.*

*Proof.* Such functions instantiate the hypotheses of Theorem 7.3, whose conclusion is exactly the desired no-draw statement and, by conjugacy, optimal termination in the original game.  $\square$

For a hostile-audit-ready completion, the refinement should expose at least the following data on every transition:

1. the current legal game states and their certified outcomes;
2. the actual draw child selected by Proposition 2.4;

3. the identities of retained token occurrences before and after the transition;
4. the exact arithmetic guard and proof that all alternatives are covered;
5. the first coordinate of  $\rho$  that strictly decreases, or the verified equal-rank control edge;
6. a finite bound or structural recursion proving that the macro returns.

This format would make the two defects in Section 8 impossible to hide: an absent parent token cannot be named in item 3, and a winning state cannot be supplied as the draw child in item 2.

## 11 Conclusion

The exact binary conjugacy and the finite-outcome semantics isolate the real difficulty of the two-player  $3n \pm 1$  game. The contracting branch, the Gray-code transducer, finite arithmetic checks, and a well-founded abstract macro rank are all useful and rigorously established. They do not by themselves show that every draw continuation enters that rank system.

The current global proof attempt fails at a specific semantic link: it uses high-return witnesses that have not been installed as retained tokens. The earlier v6.3 attempt additionally routed one loss-sibling orientation through a winning state rather than the actual draw. Accordingly, the unconditional theorem remains OPEN TARGET, NOT CLAIMED. Theorem 7.3 states a complete and machine-friendly criterion whose future instantiation would finish the problem without any further appeal to informal global induction.

## A Adversarial-audit checklist

Before a future version claims the global target as PROVED, each answer below must be supported by a proof object, a theorem reference, or an explicit finite case table with symbolic coverage.

- A1.** Is every use of “draw has a draw child” attached to the actual child selected in the next macro state?
- A2.** Is every token decreased by identity, not only by comparing numerical heights of unrelated finite states?
- A3.** When a seed is installed, which older rank coordinate has already decreased and pays for the reset?
- A4.** Are source and coefficient kept distinct under every  $3^k$  factorization?
- A5.** Are all suffix lengths handled symbolically, rather than by a fixed modulus or search depth?
- A6.** Does every loss-sibling orientation have an explicit constructor?
- A7.** Are the short exponents 1, 2, 3 proved separately wherever the generic descent uses  $D \geq 3$  or  $D \geq 4$ ?
- A8.** Does every macro return after finitely many steps, including exceptional and source-zero branches?



- A9.** Are equal-rank edges acyclic after preserving the actual full token multisets, not merely their numerical ranks?
- A10.** Has the concrete refinement, rather than only the abstract rank theorem, been checked by the proof assistant?

## References

- [1] R. K. Guy, *John Isbell's Game of Beanstalk and John Conway's Game of Beans-Don't-Talk*, Mathematics Magazine **59** (1986), 259–269. <https://doi.org/10.1080/0025570X.1986.11977258>.
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- [3] I. Althöfer, M. Hartisch, and T. Zipproth, *Analysis of a Collatz Game and Other Variants of the  $3n+1$  Problem*, Advances in Computer Games (2024), 123–132. [https://doi.org/10.1007/978-3-031-54968-7\\_11](https://doi.org/10.1007/978-3-031-54968-7_11).
- [4] G. Pochuev, *Optimal  $3n \pm 1$  Game Repository*, source code and proof notes, 2026. <https://github.com/Grisha-Pochuev/3n-plus-minus-1-game>.